

Sur Shah

925-699-5844 | sushah@ucsd.edu | [LinkedIn](#)

EDUCATION

University of California - San Diego

Master of Science in Computer Science

Bachelor of Science in Computer Science, GPA: 3.746

La Jolla, CA

Expected 06/2028

09/2023 - 06/2027

Relevant Coursework: Software Engineering Principles, Advanced Data Structures, Analysis of Algorithms, Deep and Reinforcement Learning, Operating Systems, Parallel Computing, Recommender Systems with AI, Large Codebases, Databases

SKILLS & INTERESTS

Languages: Python, Java, JavaScript, TypeScript, SQL, C, C++, HTML

Frameworks/Libraries: React, React Native, Express.js, Node.js, PostgreSQL, Firebase, PyTorch, FastAPI, Vite,

Developer Tools: Codex, Claude Code, Cursor, Git, GitLab CI/CD, Docker, Kubernetes (EKS)

EXPERIENCE

Attentive – AI Intern (Software Engineering)

Jun. 2026 – Aug. 2026

- Engineered a fault-tolerant Python runtime for multi-hour agent workflows that survives worker crashes, reaping orphaned subprocesses that escape their parent and checkpointing state so interrupted runs pick up where they left off.
- Designed a concurrency-safe state engine using SHA-256 content-addressed records and CAS commits, sustaining 2,600 writes/sec across 20 parallel workers while preventing race conditions.
- Traced a control-flow bug in a Java webhook dispatch service where a missing early return let disabled destinations still receive live payloads, and landed a fix with regression tests asserting zero outbound calls on the disabled path.

E. & J. Gallo – Software Engineering Intern

Jun. 2025 – Aug. 2025

- Architected and shipped an end-to-end license entity matching pipeline matching 2.56M+ accounts to 870K state licenses using TypeScript, SQL, and AWS Bedrock, replacing manual review and increasing compliance accuracy by **40%**.
- Built a fuzzy-matching engine with address normalization delivering **95%** precision on noisy customer data.
- Containerized the pipeline with Docker and deployed to Kubernetes via weekly cron jobs for scalable refreshes.
- Reduced cloud processing costs by **20%** by profiling redundant checks and optimizing I/O queries for parallel execution.

Triton Software Engineering – Full Stack Developer

Nov. 2024 – Jun. 2025

- Launched SpayLA's nonprofit web platform using React, TypeScript, and Firebase, enabling **100+** Los Angeles pet owners to access affordable clinic listings and educational resources.
- Increased site performance and engagement by **35%** through frontend refactoring, and optimized Firebase image delivery.
- Implemented accessibility and form validation features improving usability and trust for community users on desktop.

AlgoVerse – AI Researcher – Publication

Jun. 2024 – Nov. 2024

- Researched sarcasm detection in LLMs; designed Pragmatic Metacognitive Prompting (PMP) integrating pragmatic reasoning and contextual analysis to improve nuanced language understanding in large models.
- Implemented multi-stage prompt pipelines leveraging metacognition to outperform previous baselines by 10%.
- Achieved state of the art accuracy on trusted datasets via prompt engineering and benchmarking.
- Published findings at COLING 2025, establishing PMP as a benchmark in sarcasm/sentiment analysis and advancing LLM evaluation methodology for future language understanding tasks and developments.

LEADERSHIP EXPERIENCE

Triton Software Engineering – Vice President of Operations

Jun. 2025 - Present

- Streamlined onboarding for **100+** student developers by standardizing documentation and tooling, cutting time by **50%**.
- Reduced release issues through Git feature-branch workflows, code reviews, and CI/CD integration across 7 active projects.

PROJECTS

Jamly | React Native, FastAPI, Pytest, WebSockets, Docker

- Built a real-time audio ML pipeline using librosa and Basic Pitch to extract tempo, note events, and harmonic structure for continuous pitch and timing feedback.
- Engineered the React Native/FastAPI backend with WebSockets, Postgres, asynchronous audio processing, scalable ML inference, and persistent session tracking.

Skin Lesion Analyzer | React Native, FastAPI, PyTorch, WebSockets

- Built a mobile and web app using YOLOv8 and fine-tuned MobileNet to detect and classify 7+ skin lesion conditions from live camera input.
- Engineered the real-time React Native/FastAPI pipeline with concurrent WebSocket connections and frame processing at 8+ FPS on mobile and web hardware.